UNIVERSITY GRADUATE CAPSTONE PROJECT

DATA QUALITY REPORT & STRUCTURAL INTEGRITY AUDIT

Facing the IBM HR Analytics Employee Attrition Dataset

Course Code: DAT-602 Advanced Analytics

Document Classification: Final Project Deliverable

Academic Evaluation Boundary: Faculty of Predictive Systems

1. Executive Summary

This architectural Data Quality Report provides a systematic verification of the structural soundness, missingness profiles, duplication indices, and programmatic assertions of the raw dataset prior to downstream modeling. The data audit confirms an initial matrix size of exactly 1,470 records across 35 features. The matrix exhibits exemplary database completeness with 0.00% missing value fields and 0.00% row-wise duplications. However, structural pruning identified 3 zero-variance features (EmployeeCount, Over18, StandardHours) acting as static constants, alongside high-magnitude outliers in MonthlyIncome and YearsAtCompany requiring Winsorization. The asset yields a calculated **Data Quality Score of 91.43%**.

2. Selected Metadata Profile & Core Metrics

Feature Name	Data Type	Nulls	Unique	Mean Baseline	Std Dev (σ)	Status Target
Age	Numerical	0	43	36.92 Yrs	9.14	Validated
Attrition	Categorical	0	2	No (83.9%)	--	Target Label
DailyRate	Numerical	0	886	802.48	403.51	Validated
EmployeeCount	Numerical	0	1	1.00	0.00	Drop: Constant
EmployeeNumber	Numerical	0	1470	1024.83	602.02	Drop: High Card ID
MonthlyIncome	Numerical	0	1349	$6,502.93	4707.96	Outliers Present
StandardHours	Numerical	0	1	80.00	0.00	Drop: Constant
YearsAtCompany	Numerical	0	37	7.01 Yrs	6.13	Outliers Present

3. Programmatic Data Integrity Checks

To guarantee operational truth, the verification framework evaluated four strict organizational rules:

Assertion Rule Evaluated	Mathematical Verification Statement	Violations Count	Status
Underage Employment Check	df['Age'] >= 18	0 Records	PASSED
Positive Compensation Check	df['MonthlyIncome'] > 0	0 Records	PASSED
Tenure Validation Sequence	df['YearsAtCompany'] <= df['TotalWorkingYears']	0 Records	PASSED
Primary Key Uniqueness	df['EmployeeNumber'].nunique() == 1470	0 Records	PASSED